Regime Model Operations Guide

HMM + Macro Python Pipeline · SIM Testing Phase · 2026-05-04

V1A/B · V3C · V3D · OG

9	13	11	3	36	3	3
V1A/B ACCOUNTS	V3C ACCOUNTS	V3D ACCOUNTS	OG / STANDALONE	TOTAL ACTIVE	MODE B ACTIVE	ES PROPOSED

Infrastructure — Load Once, Not Per Chart

SHARED COMPONENTS

⚠ **CRITICAL RULE:** LiveDataExporter and ValueAreaExporter must run on dedicated support charts ONLY — not on strategy tabs. Adding these to multiple charts causes duplicate file writes and data corruption. One instance per instrument, period.

● DATA EXPORTERS

- `#1` **LiveDataExporter (NQ)** — NQ 1-min support chart → Exports\NQ_1min_export.txt
- `#1` **LiveDataExporter (ES)** — ES 1-min support chart → Exports\ES_1min_export.txt
- `#1` **ValueAreaExporter (NQ)** — NQ value area chart → Exports\ValueAre_NQ.csv
- `#1` **ValueAreaExporter (ES)** — ES value area chart → Exports\ValueAre_ES.csv
- `#1` **OFAxisMarginalVolumeExp** — V1A ES Range Volumetric chart → Active\Footprint_Export.csv
  ⚠ ONLY ONE OFAxisMarginalVolumeExp ACROSS V1A ES/NQ RUN-ACTIVE

● V3D HUD · PYTHON

- `#1` RegimeMatrixHUD_V3C — NQ leader chart → serves all NQ V3C tabs via Instrument3C
- `#17` **RegimeMatrixHUD_V3C (ES)** — ES leader chart — only when ES V3C tests are active
- `BAT` V3C_START.bat — Launches: ModelFeed Watchdog + RegimeMatrixSupervisor → keep windows open

● V3D HUD · PYTHON

- RegimeMatrixHUD_V3D — NQ leader chart → serves all NQ V3D tabs via Instrument3D
- `#17` **RegimeMatrixHUD_V3D (ES)** — ES leader chart — only when ES V3D tests are active
- `BAT` V3D_START.bat — Launches: Stage A (Macro) → Stage B (HMM) → Stage C (Supervisor) → keep all three windows open

● V1A/B SHARED

- `LIB` **HUDMessenger + HUDMessengerV1B** — Compile once — available in indicator list for all V1A/B tabs
- `LOG` **TradeLogExporter_V1AB** — One per V1A/B strategy tab. ModelVersion: V1A (Mode A), V1B (Mode B)
- `NONE` **Python Pipeline** — Not required for V1A Mode A. A+/B use shared exporters only (no Python process).

V1A / V1B — Layered Context Testing

9 ACCOUNTS · 3 STRATEGIES · 3 MODES

◆ **Mode Framework:** Each strategy runs three A/B/Mode variants on the same chart to isolate the value of each confirmation layer. `Mode A` = pure baseline (all context flags false). `Mode A+` = kill switch + daily bias (no footprint). `Mode B` = full stack including footprint signals — ⚠ PHASE 1 Footprint_Export.csv is live with history back to Jun 2024. OrderFlowSetupScanner running on V1A 15 Range Volumetric chart.

TAB NAME	ACCOUNT	MODE	CHART	SESSION	WINDOW	STATUS	NOTE
VOL FADER A – Pure	[redacted]	Mode A	3 min candle	RTH	10:35–15:45	[redacted]	No regime gates. Pure baseline.
VOL FADER A+ – Context	[redacted]	Mode A+	3 min candle	RTH	10:35–15:45	PARTIAL	Kill switch + bias. Footprint not required.
VOL FADER B – Full V1B	[redacted]	Mode B	3 min candle	RTH	10:35–15:45	[redacted]	Footprint live (Jun 2024+). Full V1B stack enabled.
KALMAN FADER A	[redacted]	Mode A	12 Range	ETH	10:35–15:45	[redacted]	Range bar normalizes tick-by-tick directional commitment.
KALMAN FADER A+	[redacted]	Mode A+	12 Range	ETH	10:35–15:45	PARTIAL	Kill switch + bias on range chart.
KALMAN FADER B	[redacted]	Mode B	12 Range	ETH	10:35–15:45	[redacted]	Footprint live. ⚠ Confirm ExitCooldownBars=3 patch deployed (17-60 incident).
COMPOSITE MOMO A	[redacted]	Mode A	5 min candle	RTH	10:35–15:45	[redacted]	DEMA(100) + 8.3tr context window.
COMPOSITE MOMO A+	[redacted]	Mode A+	5 min candle	RTH	10:35–15:45	PARTIAL	Kill switch + bias active.
COMPOSITE MOMO B	[redacted]	Mode B	5 min candle	RTH	10:35–15:45	[redacted]	Footprint live (Jun 2024+). Full V1B stack enabled.

V3C — Anchored HMM Supervisor (Shadow Model)

13 ACCOUNTS · NQ RTH ONLY

ℹ **V3C Pipeline:** Macro freshness fix now deployed. All gates check HMM state age via asymmetric hysteresis. V3C Fader 3A uses legacy macro CSV (V3_Value_Fader); 3B uses V3C HUD. This is an intentional regime-source A/B.

LANE	TAB NAME	ACCOUNT	A/B	CHART	TARGET REGIME	SESSION	STATUS
1	**Expansion A**	[redacted]	A	Unl 40-80-120	EXPAN_EXPANSION	RTH	[redacted]
1	**Expansion B 60**	[redacted]	B	Unl 60-120-180	EXPAN_EXPANSION	RTH	[redacted]
2	**Compression Sniper A**	[redacted]	A	Unl 20-40-60	TREND_COMPRESSION	RTH	[redacted]
2	**Compression Sniper B**	[redacted]	B	Unl 16-32-48	TREND_COMPRESSION	RTH	[redacted]
2	**Compression Sniper C**	[redacted]	—	Unl 20-40-60	TREND_COMPRESSION	RTH	RENAME TPL
3	**Value Fader A** (legacy CSV)	[redacted]	A	5 min candle	ROTATION_LIQUID	RTH	[redacted]
3	**Value Fader B** (V3C HUD)	[redacted]	B	3 min candle	ROTATION_LIQUID	RTH	[redacted]
4	**Momentum Slope A** (Hysteresis)	[redacted]	A	5 min candle	TREND_COMPRESSION	RTH	[redacted]
4	**Momentum Slope B** (Simple)	[redacted]	B	5 min candle	TREND_COMPRESSION	RTH	[redacted]
OG	**Momo 5m HA OG**	[redacted]	—	1 min HA	BASELINE	RTH	OG BASELINE
OG	**Momo 5m HA OG**	[redacted]	—	5 min HA	BASELINE	RTH	OG BASELINE
5	**ADX 5m Trinity OFF**	[redacted]	A	5 min candle	ROTATION_LIQUID	RTH	[redacted]
5	**ADX 5m Trinity ON**	[redacted]	B	5 min candle	ROTATION_LIQUID	RTH	FIX TYPO

V3C note: 2C template name duplicates 2A — rename to "NQ V3C Compression Sniper C 20-40-60". SB template has typo "Trinity"→ "Trinity".

V3D — Production Candidate (Python-Centralized Gates)

11 ACCOUNTS · NQ RTH + ETH (ADX DI)

✓ **Phase One Changes Deployed:** Asymmetric hysteresis gate (RC#1), macro freshness fix (RC#2), HMM feature standardization + label confidence guard (RC#3) all live. Directional bots blocked on HMM+Transition per HYST logic. SizePct locked to binary until phase-two confirms confidence-to-expectancy linearity.

SLOT	TAB NAME	ACCOUNT	A/B	CHART	TARGET REGIME	SESSION	STATUS
Top 1	**Expansion A**	[redacted]	A	Unl 40-80-120	EXPAN_EXPANSION	RTH	[redacted]
Bot 1	**Expansion B**	[redacted]	B	Unl 60-120-180	EXPAN_EXPANSION	RTH	[redacted]
Top 2	**Momentum A**	[redacted]	A	1 min candle	TREND_COMPRESSION	RTH	[redacted]
Bot 2	**Momentum B**	[redacted]	B	3 min candle	TREND_COMPRESSION	RTH	[redacted]
Top 3	**Fader A**	[redacted]	A	3 min candle	ROTATION_LIQUID	RTH	[redacted]
Bot 3	**Fader B**	[redacted]	B	5 min candle	ROTATION_LIQUID	RTH	[redacted]
Top 4	**Sniper Pullbacks A**	[redacted]	A	3 min candle	TREND_COMPRESSION	RTH	[redacted]
Bot 4	**Sniper Pullbacks B**	[redacted]	B	3 min candle	TREND_COMPRESSION	RTH	[redacted]
Top 5	**ADX DI A** (3-min, hybrid)	[redacted]	A	3 min candle	ROTATION_LIQUID	ETH	[redacted]
Bot 5	**ADX DI B** (UniRenko, hybrid)	[redacted]	B	Unl 30-60-90	ROTATION_LIQUID	ETH	[redacted]
Bot 6	**ADX DI C** (3-min, unified GuS)	[redacted]	—	3 min candle	ROTATION_LIQUID	ETH	[redacted]

ETH (S4, S8, S3): crude-level time filter (BlockOpenEnd 09:39, lunch block, SessionEnd 16:55) restricts entries to RTH windows. ETH required for Bollinger/ATR overnight bracket context.

Standalone / OG Baseline

3 ACCOUNTS · NO REGIME GATES

ACCOUNT	STRATEGY	CHART	SESSION	PURPOSE	STATUS
[redacted]	MomentumRange_RangeBar	15 Range VOL	RTH	Undocumented. Write spec or decommission.	NO SPEC
[redacted]	MomentumSlopeStrategy v1.2	1 min HA	RTH	OG baseline. Control group vs regime models.	BASELINE
[redacted]	MomentumSlopeStrategy v1.3	5 min HA	RTH	OG baseline. Template says v1.2 but strategy is v1.3 — verify.	VERSION ?

Start-of-Day Preflight Checklist

PRE-MARKET · VERIFY BEFORE 09:25

● STEP 1 — PYTHON PROCESSES

- ☐ V3D_START.bat launched — 3 windows open
  Stage A · Stage B · Stage C all running
- ☐ V3C_START.bat launched — 2 windows open
  ModelFeed Watchdog · RegimeMatrixSupervisor
- ☐ No Python errors in console windows
  Look for "CRITICAL" or "MACRO_STALE_WRONG_DESIGN" on first run

● STEP 2 — DATA EXPORTERS (SUPPORT CHARTS)

- ☐ NQ 1-min support chart loaded
  LiveDataExporter active → NQ_1min_export.txt updating
- ☐ ES 1-min support chart loaded
  LiveDataExporter active → ES_1min_export.txt updating
- ☐ ValueAreaExporter running (NQ + ES)
  ValueAre_NQ.csv and ValueAre_ES.csv updated today
- ☐ LiveDataExporter NOT on strategy tabs
  Only 1 instance per instrument. Check NTB strategy list

● STEP 3 — HUDS (LEADER CHARTS)

- ☐ RegimeMatrixHUD_V3D on NQ leader chart
  V3D\NQ_RegimeMatrix_Latest.csv → IsStaleDailyFlag = False
- ☐ RegimeMatrixHUD_V3C on NQ leader chart
  V3C\NQ_Regimes_V3C_latest.csv → HUD fixed
- ☐ HUDMessenger + HUDMessengerV1B compiled
  Available in indicator list. Not a per-chart item.

● STEP 4 — V1B CONTEXT (A+/B ONLY)

- ☐ Daily bias fresh on Mode A+ tabs
  TradeHUD shows today's DIR/b/B bias
- ☐ Footprint_Export.csv status
  ⚠ PHASE 1 ACTIVE
  Verify: ls -lh Active\Footprint_Export.csv (Jun 12 Range chart feeds today's row, not appending)

● STEP 5 — STRATEGY ACCOUNTS

- ☐ All 34 accounts enabled in NTB
  Check Control Center → Account list
- ☐ Account names match registry exactly
  Sim101-NQ-SC must be separate from SB
- ☐ Template matches account on every tab
  No V3D template on a V3C account vc vice versa

● STEP 6 — FIRST CHECKPOINT VERIFY

- ☐ At 09:35: HUD shows live FinalRegime
  Not TRANSITION-only · Check AsymHysteresisReason in CSV
- ☐ StateReason = FRESH on all HUDs
  If MACRO_STALE_WRONG_DESIGN → yesterday's data not flushed
- ☐ LabelAmbiguousFit = 0 on HMM output
  If 1 → HMM centroid collapse. Check HMM console.

End-of-Day Checklist

POST 16:00

● TRADE LOG VERIFICATION

- ☐ V3D trade rows stamped model_version=V3D
  No V3C accounts in V3D log
- ☐ V3C trade rows stamped V3C / Anchored_HMM_V3C
  No V3D logged on V3C tabs
- ☐ UNIFIED AllModels export generated
  Check UNIFIED\AllModels_TradeLog_YYYYMMDD.csv row count
- ☐ V3D directional lanes: no Transition trades
  Phase One gate: AllowPosInTransition=0 when HMM=Transition

● OUTPUT FILES

- ☐ V3D\NQ_RegimeMatrix_Full.csv updated
  Run --backfill if not auto-run
- ☐ V3C\NQ_Regimes_V3C.csv updated
  Check last row timestamp matches today
- ☐ AsymHysteresisReason distribution reviewed
  All HYST_DISABLED → kill-switch off. All MACRO_STALE → feed issue.

HUD Pipeline Status Upgrade — Implementation Guide

V3C HUD · NEW V1A HUD · GREEN/YELLOW/RED INDICATORS

ℹ **Goal:** Surface pipeline health (Python processes, CSV freshness, HMM label quality, footprint status) as green/yellow/red indicators visible at a glance in NTB — without requiring the operator to inspect CSV files or Python console windows. V3D's HUD already shows confidence, conflict, and regime label. This extends that pattern to V3C and adds a new lightweight V1A HUD.

● V3C HUD — Pipeline Status Panel

Add a new "Pipeline Status" section to `RegimeMatrixHUD_V3C.cs` that reads the same CSV files the strategies consume and renders colored indicators.

1. Add read of `NQ_Regimes_V3C_latest.csv` timestamp. If file age > 7 min = YELLOW; > 15 min = RED.
   `// Pipeline health: check CSV freshness`
   `var csvAge = DateTime.Now - File.GetLastWriteTime(csvPath);`
2. Read `LabelAmbiguousFit` from HMM CSV. If = 1, show YELLOW "HMM DRIFT" warning.
3. Read `AsymHysteresisReason`. Parse HYST_BLOCK vs HYST_PASSED for current row. Show "GATE: OPEN ●" in green or "GATE: BLOCKED ●" in red.
4. Check if `NQ_1min_export.txt` exists and was modified within last 3 min. GREEN = live feed; RED = feed down.
5. Render a compact status strip below the regime label panel. Use `Draw.TextFixed` with background color matching status. One line per indicator. No popups.

● V3D HUD — Additions Only

V3D HUD already shows regime, confidence, direction, and conflict. Add three new fields from the Phase One changes:

1. Show `AsymHysteresisReason` as a compact "HYST" line. Color: green = PASSED, red = BLOCKED, gray = DISABLED.
   `// Phase One: show hysteresis gate status`
   `string hstr = (bar == MyAsymHysteresis.LastBar) ? MyAsymHysteresis.LastReason : "";`
2. Show `HMMStateAgeBars` as "HMM AGE: N bars". YELLOW if < 2 (below persistence threshold).
3. Show `LabelAmbiguousFit` from HMM CSV. If = 1, flash "⚠ HMM DRIFT" in yellow background.
4. Show Python Stage A/B/C output file ages in a "PIPELINE" line: each stage file age < 7 min = ● green; else ● red.

● NEW: V1A Pipeline Monitor HUD

A new lightweight indicator `PipelineMonitor_V1A.cs` for V1A/B charts. No regime logic — only health indicators.

1. Check `Footprint_Export.csv` last-modified timestamp and row count. Green = rows present + modified today. YELLOW = rows present but > 1 session old (stale). RED = file missing or 0 data rows. History exists back to Jun 2024 — any RED here means the OrderFlowSetupScanner stopped writing.
2. Check `NQ_1min_export.txt` last-modified timestamp. < 3 min = GREEN. 3–10 min = YELLOW. > 10 min = RED "LIVE FEED DOWN".
3. Check `ValueAreaExp_NQ.csv` exists and contains today's date. GREEN = today. YELLOW = yesterday. RED = missing.
4. Read daily bias from HUDMessenger (if accessible). Show current bias label: D / P / b / B. GRAY if Mode A (no bias used).
5. Render as a 4-line vertical strip, top-left of chart. Each line: ● [color] LABEL. One per check. Compact, always visible. Add to any V1A/B leader chart — does not need to be on every tab.

◆ **Pipeline Status Mockup — What Each HUD Should Display:**

```
PIPELINE STATUS
● LIVE FEED (NQ 1-min)      LIVE · 0:07 ago
● HMM WATCHDOG              RUNNING
● MACRO SUPERVISOR          FRESH · 0:13 ago
● HMM LABEL QUALITY         TRN TRANSITION
● FOOTPRINT FEED            LIVE · 1m 00s
● ASYM GATE                 PASSED "PASSED"
● VALUE AREA                Fvgap · 4m 5s
```

This visual layout can be implemented in any NinjaScript indicator using Draw.TextFixed with dynamic background brush selection based on each status value.

Open Action Items

PRIORITY RANKED

PRIORITY	ITEM	ACCOUNTS / FILES	EFFORT
P1	**Create dedicated Sim101-NQ-SC account** — SB currently shares ADX_DI_V3D (UniRenko) + ADX_DI_V3D_C (3-min GuS). Two strategies on one account contaminates trade attribution and makes the SA vs SC hybrid-vs-unified GuS comparison unmeasurable.	Sim101-NQ-SB/SC	New SIM account
P2	**Confirm ExitCooldownBars=3 patch deployed** on Sim1A Kalman Fader 3 (Mode B). This account had the 17-consecutive-fill incident. Mode B is now active with live footprint data — **must verify patch is in place before this account trades.**	Sim1A Kalman Fader 3	Inspect — verify before trading
P3	**Rename V3C Compression 2C template** from "NQ VC3-A Compression Fast 20-40-60" (duplicate of 2A) to "NQ V3C Compression Sniper C 20-40-60".	Sim1V3C-NQ-2C	Template change
P4	**Fix template typo** in V3C ADX SB: "Trinity" → "Trinity"	Sim1V3C-NQ-SB	Template rename
P5	**Implement HUD pipeline status indicators** — V3D additions (AsymHysteresisReason, HMMStateAgeBars, LabelAmbiguousFit, stage file ages), V3C additions (state reason, hysteresis gate), new V1A PipelineMonitor_V1A.cs. See engineer handoff document.	All HUDs	C# indicator work
P6	**Add 3 ES V3D accounts**: ES Expansion (Unl 22-44-66 RTH), ES Fader (1-min RTH), ES Sniper (3-min RTH). Validates ES-specific asymmetric hysteresis gate rules (ES Balance fortress, ES Trend(b) quarantine).	New accounts	New setup × 3
P7	**Verify SimMomo2 template version** — template name says v1.2, strategy is v1.3. Align to avoid logging confusion in unified trade log.	SimMomo2	Inspection only
P8	**Clarify Sim1 chart relationship** — MomentumRange_RangeBar runs on 15 Range VOL. OrderFlowSetupScanner is on 15 Range chart. Verify these are separate chart instances or document the shared-chart arrangement. Write spec or decommission Sim1.	Sim1	Decision + optional doc
P9	**Phase Two trade-log cross-tabs** once uniform exports land. Run win-rate/expectancy by AsymHysteresisReason, per-state profit factor, V1B Mode A+ vs B comparison (footprint now live = first empirical test), confidence-expectancy linearity test for SizePct validation.	All accounts	Phase Two analysis